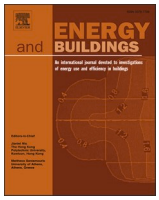
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Monitoring indoor environmental conditions in office buildings using a sustainable Agentic RAG-LLM system

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ABSTRACT

Indoor Environmental Conditions (IEC) play a crucial role in determining the health, productivity, and overall building performance of employees, as well as their energy consumption. Key parameters, such as temperature and humidity, are not only vital for thermal comfort but also offer opportunities to enhance energy efficiency when effectively monitored and managed. Accurate Thermal Comfort Monitoring (TCM) remains challenging to achieve because it requires the integration of diverse data sources and intelligent analysis, particularly in light of evolving global energy and sustainability standards. Although Building Information Modeling (BIM) is increasingly being adopted to manage complex building data, its integration with real-time sensor inputs remains vastly underutilized. Existing thermal monitoring systems are often development-intensive, require significant domain expertise, lack Natural Language (NL) interaction capabilities, and are not inherently adaptable, necessitating frequent technical upgrades. These limitations give rise to pressing concerns about long-term scalability, usability, and sustainability. To address these limitations, this study introduces *ThermalComfortBot*, an integrated Information System (IS) powered by Generative Artificial Intelligence (GenAI). *ThermalComfortBot* utilizes open-source technologies, including Large Language Models (LLMs) and Agentic Retrieval-Augmented Generation (RAG), to enhance thermal comfort and support energy optimization in buildings. The system integrates Building Information Modeling (BIM), sensor data, and external datasets to generate actionable insights, delivered through both textual explanations and graphical visualizations. This system utilizes flexible and adjustable LLMs that are guided by principles of sustainability, thereby making them cost-efficient, scalable, and practical for a diverse range of organizational environments. In a real-world case study, *ThermalComfortBot* outperforms traditional RAG-LLM, achieving 94% accuracy, 92% precision, and 89% recall, enhancing comfort and efficiency.

1. Introduction

IECs in the workplace have a significant influence on employees' physiological and psychological well-being [1–4]. These conditions can lead to changes in habits, well-being, and physical and cognitive productivity [4,5]. Temperature [6] and humidity [7] are among the various factors that affect indoor environmental quality and thermal comfort [8,9]. The UK Health and Safety Executive (HSE) defines thermal comfort as the result of a combination of environmental, work-related, and personal factors that jointly determine a comfortable workplace temperature [10]. Research [7,11,12] has consistently shown that maintaining optimal thermal comfort is crucial for assisting high productivity levels. For instance, existing studies [13–15] have shown that productivity peaks when the office temperature is around 21 °C. On the contrary, employees tend to feel fatigued in overly warm workplaces, whereas overly cold environments can cause restlessness and distraction [6]. Humidity also plays a key role in achieving thermal comfort. The HSE recommends maintaining workplace Relative Humidity (RH) levels between 40 % and 70 % [10]. Humidity levels below 30 % can cause discomfort, which includes dry eyes and skin, whereas levels above 70 % can lead to a stuffy environment [6]. Research [6,16,17] has consistently shown that actual comfort is only attained when both temperature and humidity fall within this "comfort zone". TCM plays a crucial role not only in supporting health and productivity but also in offering potential to improve energy efficiency standards. Standardized seasonal assessments are essential for ensuring optimal conditions throughout the year.

Data collection and evaluation of energy and environmental

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performance are central to the implementation of effective building energy efficiency strategies [18]. Environmental monitoring technologies play a crucial role in these efforts [18]. Additionally, there has been a global increase in the number of energy-related building legislation and regulations [19,20]. For instance, the European Energy Performance of Buildings Directive (EPBD) promotes the use of intelligent metering and active control systems in buildings through regulatory measures [21]. BIM has emerged as a powerful tool for managing building-related data in the construction industry [22]. However, its application in IECs remains limited, as its integration with sensor data has yet to be fully explored. Compared with traditional thermal condition monitoring systems, integrating BIM models with sensor data can significantly enhance real-time monitoring, providing owners, facility managers, and occupants with immediate, actionable insights and a more user-friendly interface. While numerous systems [23–34] have been developed to integrate BIM with sensor technology, most are proprietary, and the few open-source options often lack the flexibility to adapt to dynamic environmental conditions. These systems struggle to accommodate evolving IS requirements, such as the addition or removal of building-associated sensor data. This lack of adaptability makes it challenging to evaluate, compare, and optimize these systems. Furthermore, the time and resources required to assess proprietary solutions individually can result in inefficiency and increased costs.

The absence of an integrated, dynamic, and open-source system hampers the effective aggregation and utilization of information, further complicating efforts to enhance the energy efficiency and performance monitoring of indoor building environments. To address these challenges, this study proposes *ThermalComfortBot,* an integrated IS that functions as a Chatbot, designed to synthesize data from diverse sources for thermal comfort analysis. *ThermalComfortBot* is designed explicitly for TCM in facilities management and utilizes open-source technologies, including LLMs and Agentic RAG. A key innovation in *ThermalComfortBot* lies in its use of Agentic RAG, which goes beyond passive Information Retrieval (IR). The system exhibits goal-directed behavior by leveraging an autonomous agent to plan, retrieve, and reason across heterogeneous datasets, which is essential for adaptive thermal comfort management. This dynamic approach significantly enhances contextual understanding, response relevance, and decision-making in real-time building environments. Moreover, the proposed system offers several key benefits that distinguish it from conventional RAG-LLM approaches used in other domains.

1) *Purpose-Built for Facilities Management*: *ThermalComfortBot* is uniquely designed for the specialized field of facilities management. It addresses thermal comfort optimization explicitly, with direct implications for occupant health, operational efficiency, and building energy management.
2) *Integration of Diverse Data Sources with Real-World Validation*: While existing RAG-LLM systems principally rely on textual corpora, *ThermalComfortBot* distinguishes itself by integrating structured (BIM), semi-structured (sensor data), and unstructured information to deliver timely, context-aware insights. Its effectiveness has been validated through a real-world case study, showcasing both the system's innovative approach and its practical utility in building TCM.
3) *Graphical Output Capabilities*: *ThermalComfortBot* provides graphical outputs that facilitate faster and more intuitive interpretation of building and environmental data, marking a significant enhancement in user experience and functionality. This differs from existing RAG-LLM systems, which are limited to text-based responses.
4) *Tailored for Cost-Effective Decision Support and Sustainability-Driven Design*: Engineered as a decision support system, not merely a Chatbot, *ThermalComfortBot* simplifies complex analytical workflows, reducing the cognitive burden on building analysts. Reusable LLMs offer a scalable and cost-effective approach, promoting environmental sustainability and aligning seamlessly with contemporary energy efficiency goals.

The remainder of the paper is organized as follows: Section 2 outlines the study's background and context. Section 3 introduces *ThermalComfortBot*, an innovative IS that integrates Agentic RAG with LLMs. Section 4 details the testing of the proposed system through a case study. The advantages and limitations of *ThermalComfortBot* are discussed in Section 5. Finally, Section 6 presents concluding remarks and implications for the study.

2. Background

This section is divided into two parts. The first part offers an overview of BIM and reviews key studies on the use of BIM and sensor-based technologies for environmental monitoring. The second part introduces LLMs integrated with RAG, and discusses relevant research on their applications. Finally, a critical research gap exists, emphasizing the facilities management industry's need for more comprehensive approaches that leverage existing LLMs and RAG technologies.

2.1. Overview of BIM and existing research on BIM-Sensor integration

The Architecture, Engineering, and Construction (AEC) industry leverages BIM to enhance cost-efficiency, productivity, and project quality [23]. BIM is a powerful platform for managing complex building information and enabling virtual visualization throughout a building's lifecycle [24]. It uses digital modeling software to efficiently design and execute projects, with BIM models representing virtual equivalents of actual building sections. These models play a crucial role in evaluating energy efficiency during the design phase by considering various parameters, including orientation and materials. The real-time monitoring capabilities of BIM allow facility managers and engineers to efficiently oversee building operations by visualizing data captured by sensors, particularly for maintaining thermal comfort within predefined thresholds. Recent studies have emphasized efforts to integrate BIM with sensing technologies [24]. Some researchers have embedded sensor data in Industry Foundation Classes (IFC) formats to ensure full interoperability. Developed by the International Alliance for Interoperability (IAI), now known as buildingSMART, IFC is an open-exchange format compatible with various BIM tools, facilitating project standardization [23].

Several studies have explored the integration of BIM for environmental monitoring of buildings, each focusing on different aspects and datasets. Lin & Cheung [23] developed an environmental monitoring system for parking garages that utilizes data on gas, temperature, and humidity. The system is built using the C#.NET Framework and facilitates regular management and informed decision-making during critical situations. Desogus et al. [24] monitored building consumption and indoor conditions, using data from heat pumps, electrical energy meters, and luminance sensors. Their system, created with the Dynamo visual programming platform, provides both real-time and historical data to enhance building management. Kensek [25] integrated environmental sensors with BIM to test intelligent façade systems. By processing light, humidity, and CO2 data through Arduino, Dynamo, and the Revit Application Programming Interface (API), this approach enabled the testing of façade systems before constructing full-sized mock-ups, thereby improving design efficiency and effectiveness. Zhong et al. [26] proposed an ontology-based framework for environmental monitoring of buildings, employing semantic web technology to manage temperature, humidity, noise, light, and air quality. This framework facilitates the inference of critical information necessary for effective environmental monitoring. Opoku et al. [27] extended environmental monitoring to living laboratories and used a cloud-based digital twin system to manage sensor data.

Valinejadshoubi et al. [28] developed a TCM alert system. It utilizes

temperature and humidity data to provide a robust tool for intelligent monitoring of thermal conditions, facilitating the collection and analysis of data. Wu & Li [29] developed a system that uses temperature, humidity, and carbon dioxide data to enhance occupant comfort while efficiently operating energy-intensive equipment. COZyBIM balances comfort and energy use by integrating Autodesk Revit with PHP and SQL to achieve energy-saving targets. Natephra & Motamedi [30] focused on sensor data visualization for indoor conditions monitoring. Their system, which combines temperature, humidity, and light level data with Dynamo, Arduino, Virtual Reality (VR), Autodesk Revit, and Autodesk 3ds Max, provides TCM. Natephra et al. [31] integrated thermal images with BIM to evaluate thermal comfort in naturally ventilated environments. Using Autodesk Revit and thermographic cameras, this method offers statistical data to analyze thermal performance and identify areas that need improvement. Alavi et al. [32] employed a BIM-based probabilistic approach to enhance occupant comfort by considering factors such as thermal comfort, indoor air quality, and visual comfort. Using Dynamo, Autodesk Revit, and Python, this study helps identify the causal factors contributing to discomfort, enabling targeted interventions. Shahinmoghadam et al. [33] explored the assessment of thermal comfort in building enclosures using thermal images, cloud platforms, edge computing, and Raspberry Pi. Finally, ElArwady et al. [34] focused on modeling indoor thermal comfort in buildings using temperature and humidity data. This platform allows for the active adjustment of indoor conditions through the control of air conditioning devices, ensuring that comfort levels are efficiently maintained.

These are just a subset of the systems available in the literature on BIM-sensor integration for building TCM. Small and medium-sized enterprises (SMEs) with limited Research and Development (R&D) resources, as well as facility managers and building operators, often face challenges in identifying, evaluating, and testing existing systems, making it difficult to find tailored solutions suited to their specific needs [35]. The dynamic nature of building environments necessitates continuous updates to systems and datasets, which can demand significant time and financial investment, often making them impractical for SMEs with limited resources. Moreover, many studies [29–34] have focused on integrating BIM and sensor data, which limits the breadth and depth of available information, thereby hindering comprehensive decision-making. This approach can introduce bias and overlook valuable insights from multiple data sources, including HSE guidelines, weather data, occupancy patterns, historical maintenance records, and user feedback. Incorporating additional information sources increases the complexity of data linking and semantic enrichment, i.e., connecting and enhancing information with meaningful context [36]. Semantic enrichment is essential for integrating diverse data streams to create a holistic view of thermal comfort in buildings [36]. For instance, linking indoor temperature and humidity data with HSE guidelines ensures that sensor insights align with safety and comfort standards. Integrating BIM data enables facility managers to identify specific rooms affected by anomalies in sensor readings and facilitates targeted assessments.

Moreover, occupancy schedules and occupant feedback provide occupancy context, as relying only on sensor data overlooks how varying occupancy patterns influence comfort perceptions during the day. Facility managers and stakeholders gain a complete view of building conditions when sensor data is enriched with HSE guidelines that recommend maintaining comfortable indoor temperatures for workers by monitoring air temperature, humidity, radiant heat, and air movement. This information is augmented with BIM information, occupant feedback, and usage patterns. This approach enables specific adjustments to improve thermal comfort, boost energy efficiency, and ensure employee well-being and productivity. Employing advanced semantic enrichment methods [36] is crucial for connecting information from multiple sources based on similar contexts, fostering a thorough understanding of building thermal comfort. Rather than relying on complex semantic enrichment techniques, which can incur significant computational costs, utilizing LLMs based on Natural Language Processing (NLP) offers a sustainable alternative [37]. The technologies for creating these integrated systems should also be upgradable and customizable, minimizing both technological and labor costs. This strategy promotes sustainable systems for SMEs by encouraging responsible investment. The following section explores the potential of combining LLMs with RAG to achieve these goals.

2.2. Studies on LLMs and RAG integration

LLMs are pre-trained on vast datasets and can understand and generate text that closely mimics human language [37]. This comprehensive pre-training mechanism enables LLMs to excel in executing various NLP tasks, including text generation, language understanding, IR, text classification, sentiment analysis, and text summarization [38]. Employing deep learning techniques in LLMs enables them to identify and learn complex patterns and relationships within a language. LLMs' capabilities are only restricted to the scope of the datasets that were used during their training phase. However, RAG enables LLMs to execute various NLP tasks using private datasets. It adds an extra step in which the LLM retrieves relevant information from external datasets before generating responses [39]. The addition of the external IR process enhances the accuracy and relevance of the response generated, thereby reducing errors and improving the overall quality of LLM output.

The diverse applications of RAG across various domains highlight its flexibility in improving Question-Answering (QA) and Information Extraction (IE) tasks. For example, LiVersa [40] uses RAG for liver disease-specific data to improve diagnostics. Almanac [41] incorporates clinical guidelines to support clinical decision-making. In education, RAG enhances decision-making through applications like the Assessment of Tutoring Practices [42]. RAG's ability to handle out-of-domain scenarios [43] expands its use, addressing textbook QA in life and earth sciences. Automated Form Filling [44] simplifies IT project requests by improving form completion. In health, Frontline Health Worker Capacity Building [45] processes pregnancy-related guidelines for effective health education. Self-BioRAG [46] enhances biomedical informatics via instructional sets, improving information management in research and practice. RAG also benefits technical and humanitarian fields. For instance, RAG-Fusion [47] improves technical information QA using product datasheets, while FloodBrain [48] employs RAG for flood disaster reporting, providing appropriate support for humanitarian efforts. In scientific and traffic management applications, Clinfo.ai [49] improves medical QA, and RealGen [50] utilizes the nuScenes dataset for traffic scenario management. Code4UIE [51] develops IE and code generation capabilities. Additionally, RAP [52] enables planning and decision-making processes. In the field of human resources, the study Enhancing Multilingual Information Retrieval in Mixed HR Environments [53] utilizes RAG to optimize HR documents and multilingual QA. More recently, a study [54] introduced the Energy Chatbot, designed to bridge the information gap for SMEs in Sustainable Energy Transitions (SETs). By combining different media sources into a centralized IS, it improves decision-making, cost efficiency, and access to sustainability insights.

While numerous RAG-LLM systems have shown success in domains such as biomedical research and finance, their application in facilities management, particularly for thermal comfort optimization, remains significantly underexplored. Several key shortcomings limit their broader adoption in this context:

1) *Narrow data focus*: Most existing solutions [40–54] rely on textual data while neglecting structured (e.g., BIM) and semi-structured (e.g., real-time sensor) datasets, which are crucial for generating context-aware insights in buildings.
2) *Lack of integration*: The inability to seamlessly incorporate real-time environmental data and building layout information weakens the system's capacity to assess and optimize IECs.

3) *Text-only outputs*: Existing systems [40–54] primarily provide responses in text form, overlooking the importance of visualizations that can facilitate intuitive interpretation by users.
4) *Usability gaps*: Without visualizations and integrated data analysis, many systems fall short in supporting facilities professionals, such as building analysts and managers, who need actionable insights for effective decision-making.

To overcome these limitations, this study uses Agentic-RAG, an advancement of traditional RAG architecture. Agentic-RAG integrates an LLM with agent-based reasoning and dynamic prompt orchestration, enabling the system to manage query handling, data source routing, and context refinement [55,56]. By combining embeddings from structured, semi-structured, and unstructured sources, and returning outputs in both textual and graphical formats, Agentic-RAG significantly enhances the accuracy, adaptability, and usability of the system. This architecture supports the development of *ThermalComfortBot*, a domain-specific solution designed to improve thermal comfort in buildings through context-aware and user-friendly NL interactions.

3. Proposed ThermalComfortBot System

This section provides an overview of *ThermalComfortBot*, an LLM-powered Agentic-RAG system, designed as an IS for monitoring building thermal comfort. It starts by offering the overall concept of the system. Next, it outlines the key data sources essential for implementing this integrated system. Furthermore, this section outlines the system architecture, showing the implementation of the Agentic RAG and LLM-based *ThermalComfortBot* for efficient IR, processing, and decision-making in monitoring building thermal comfort.

3.1. System overview

The *ThermalComfortBot* system for building TCM is designed to assist building and facility managers in making informed and data-driven decisions about thermal comfort and, as a result, taking remedial actions. The system leverages Agentic-RAG and LLM technologies to provide insights based on a combination of current and historical data [57]. Fig. 1 presents a high-level view of user interaction with the *ThermalComfortBot* system, indicating how the system operates in a real-world scenario. The workflow is structured into three main stages: input, processing, and output.

Input Stage: The process begins when a Building Analyst submits a query through the *ThermalComfortBot* interface, a Chatbot accessed via a simple text box labeled *"Enter your question here"*. The Chatbot accepts NL input utilizing the advanced capabilities of LLMs and can accurately interpret queries even when they are not grammatically correct. This functionality ensures that users do not require technical expertise in ISs. They only need the basic ability to frame and type a question. The system is designed to respond to queries that fall within its predefined scope, mainly focused on issues related to building thermal comfort. However, this scope is configurable. The system can be tailored to include or exclude specific topics such as user feedback, weather data, or other building performance indicators. This flexibility enhances accessibility and usability by allowing the system to accommodate a wide range of analytical needs and user preferences, regardless of technical background.
Processing Stage: The system uses a combination of Agentic RAG and LLM components to process the query. While many LLMs are available, such as GPT [58], PaLM [59], and LLaMA [60], LLaMA was selected for this implementation due to its open-source nature and showed strong performance in existing studies [60]. However, developers have the flexibility to choose and integrate any other LLM that aligns with their specific requirements. The RAG scans relevant data sources for the most contextually appropriate information. These sources include BIM data, sensor data (such as temperature and humidity readings), weather data, historical maintenance records of sensors, HSE guidelines, occupancy data, user feedback on thermal comfort, and Building Analysts' reports. The LLM then synthesizes this information, generating a coherent and comprehensive response. Data formats like IFC, comma-separated values

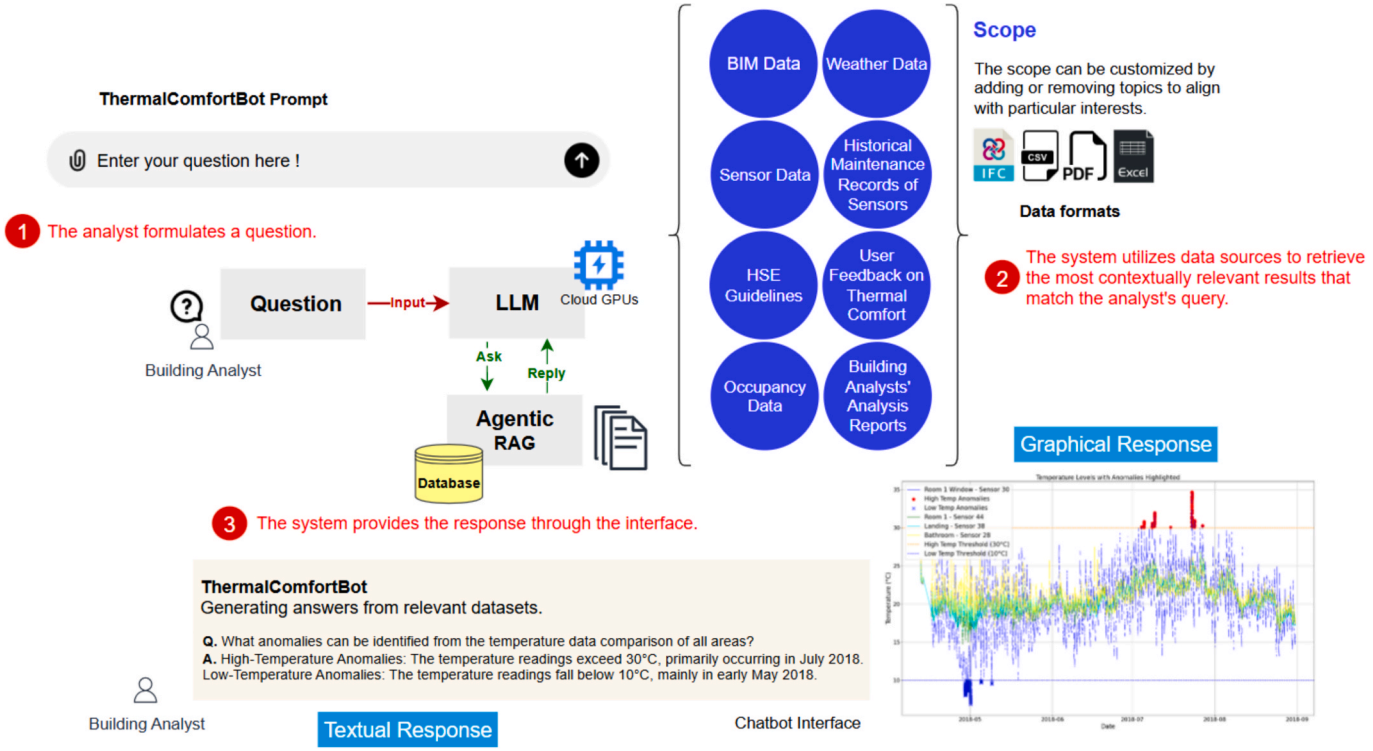


Fig. 1. The building analyst interacts with *ThermalComfortBot*, which uses diverse datasets in various formats to respond to queries.

(CSVs), PDF, and Excel are supported, ensuring a diverse dataset for analysis, which supports the depth of the system's response.
Output Stage: After processing the input, the system generates a context-aware textual response, along with a corresponding graphical output, displayed directly through the Chatbot interface. This user-friendly presentation enables building analysts to easily interpret the results and make informed decisions related to IECs, regardless of their technical background.

The system is designed to simplify complex processes, such as data aggregation, semantic enrichment, and analysis, thereby significantly enhancing operational efficiency in building TCM.

3.2. Relevant datasets

RAG-integrated LLM systems rely on the datasets that drive them [57]. Therefore, the quality of their response is primarily linked to the completeness and reliability of the data provided. Fig. 2 illustrates the essential datasets critical [28–35] for building TCM. These data types are categorized by their nature and function, creating a comprehensive framework for understanding the factors affecting thermal comfort in buildings. BIM data provides structural details about a building's design, adhering to the IFC standard. This information helps stakeholders understand the physical environment [23–25]. Sensor data offers time-series insights into temperature and humidity levels over time, providing information about the indoor climate. Weather data provides information on external environmental factors, such as temperature and humidity, all of which significantly impact indoor thermal comfort. HSE Guidelines [10] offer regulatory insights into health, safety, and best practices in building comfort management. Occupancy data reveals usage patterns affecting air quality and temperature, while maintenance records of deployed sensors ensure sensor reliability. User feedback provides valuable insights into comfort levels, while analysts' reports offer expert recommendations. Together, these datasets enable a comprehensive understanding of thermal comfort. This list is not complete. Additional data sources can be added or excluded based on the system's goals. With these datasets defined, the Agentic RAG-integrated LLM system is developed as described in the next section.

3.3. System architecture

The system architecture (see Fig. 3) enables the functionality of the *ThermalComfortBot*, as described in Fig. 1. This architecture outlines the technical workflow that powers the *ThermalComfortBot*, involving a series of interconnected steps that ensure efficient IR, embedding storage, and LLM-based querying. The architecture is divided into seven stages, each playing a critical role in processing and retrieving relevant data:

1. *Install Data Processing Libraries*: Key libraries include Langchain [61], Langchain-Community, Sentence-Transformers [62,63], and Facebook AI Similarity Search (FAISS) [64] for Graphics Processing Unit (GPU). These libraries enable the handling of data processing, embeddings, and fast retrieval mechanisms, laying the foundation for the system.
2. *Load and Prepare the Data*: After configuring the necessary libraries, structured data (e.g., BIM file), semi-structured data (e.g., sensor logs), and unstructured data (e.g., weather reports, HSE guidelines) are loaded and prepared for embeddings. The diversity of datasets ensures the system can provide context-aware insights by correlating different information sources. The process reads data [65] and utilizes the *CharacterTextSplitter* to break the text into chunks. This is a crucial step to ensure that the data is properly structured for subsequent embeddings and processing.
3. *Load and Apply Embeddings*: Embeddings are applied to the data chunks using the Hugging Face embedding model (i.e., huggingface.co), with the model running on the GPU for efficient performance. Since the system deals with various data formats (IFC files, CSV files, and text files), different embeddings are applied depending on the data type. This ensures that the textual, numerical, and structural information from each dataset is accurately converted into vector representations, making it convenient for the system to understand and retrieve relevant information in the subsequent stages.
4. *Store Embeddings with FAISS*: FAISS [64] is used to store embeddings of data chunks that were generated. It allows fast similarity searches across large datasets, which is crucial for the IR. The FAISS index is saved locally, allowing it to be reloaded when needed. This index serves as a repository for all embedded data that is prepared for querying at any time.
5. *Setting Up the LLM Model*: The next step involves setting up the LLM (i.e., LLaMA 3.1 [60] in our case) using the *Langchain-Ollama* package [66]. The *Ollama* Environment is instantiated, and the system integrates the LLM with the embeddings stored in FAISS. This allows the LLM to effectively retrieve and process information from the

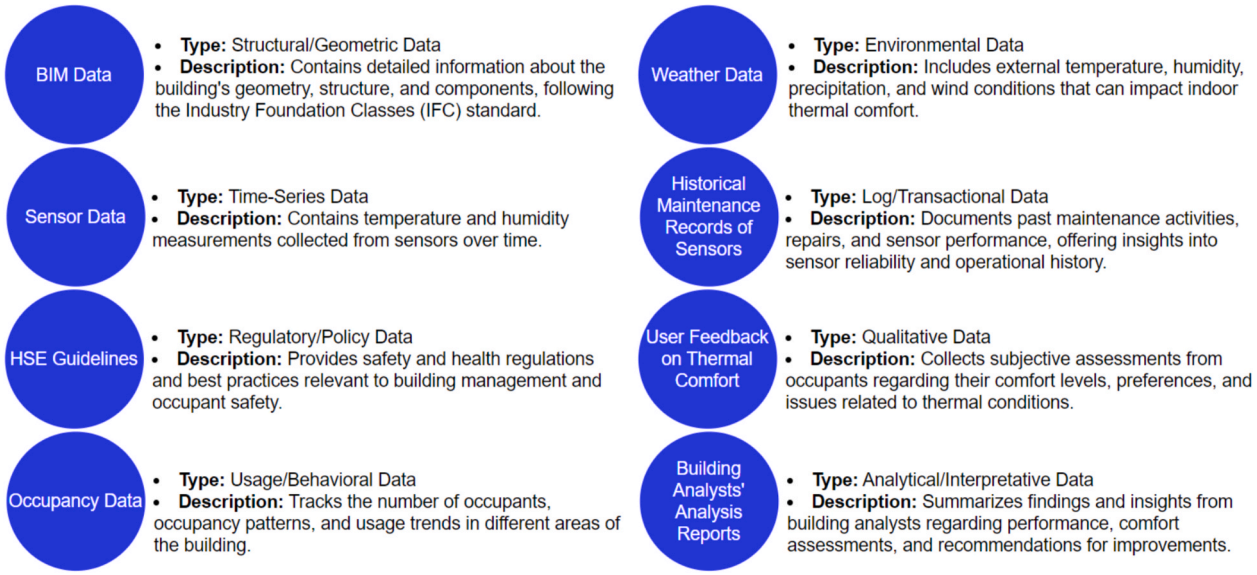


Fig. 2. A selection of key datasets for understanding building thermal comfort.

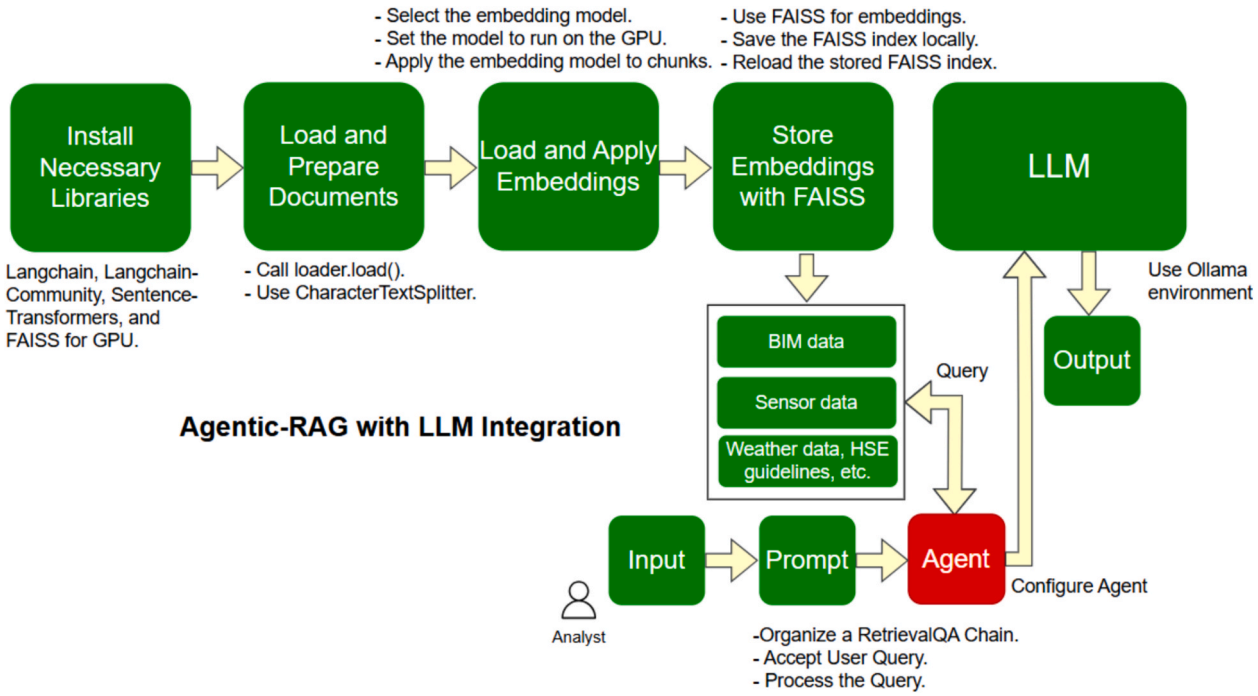


Fig. 3. System architecture for developing an Agentic RAG with an LLM-powered *ThermalComfortBot*.

embedded data, ensuring that it can generate accurate and contextually relevant responses.

6. *Configure the Agent and Prompting Logic*: An agent is configured to manage the interaction between user queries, the LLM, and the FAISS index. This includes organizing a *RetrievalQA Chain* [67], processing inputs, and constructing prompts that maximize the relevance and clarity of the LLM's response.
7. *Handle User Input and Generate Output*: The user (analyst) provides a query through an NL interface. The agent processes this query, retrieves the most contextually appropriate data from the FAISS index, and feeds it to the LLM. The output is generated in both textual and graphical formats to support intuitive understanding and decision-making.

Each stage of the system plays a crucial role in ensuring that *ThermalComfortBot* delivers accurate and relevant insights for optimizing building TCM. To handle queries that involve cross-referencing multiple datasets, the system utilizes the aforementioned RAG implementation stages, which seamlessly integrate different datasets. Upon receiving a user query, the agent identifies the relevant data types, retrieves contextually appropriate information using embedding-based similarity search and rule-based logic, and combines the results. This integrated data is then passed to the language model, which generates a context-aware response. This mechanism enables the system to address complex queries involving spatial, temporal, and semantic relationships, such as detecting thermal anomalies in specific zones or connecting sensor patterns with architectural layouts. The following section presents a detailed case study demonstrating how *ThermalComfortBot* operates in a real-world building environment, showcasing its practical application in delivering actionable insights for thermal comfort optimization.

4. Case study

4.1. Overview

This section presents a real-world case study demonstrating *ThermalComfortBot's* application using diverse datasets to improve occupant thermal comfort through conversational NL interactions. The case study is set in a two-story stone structure constructed using traditional limestone masonry, specifically Cotswold stone, a common material in rural England. The construction adheres to regional architectural practices, featuring thick stone walls that provide high thermal mass and durability. The internal building layout, as shown in the floor plan (see Fig. 4), comprises three rooms: Room 1 in the eastern wing, Room 2, centrally located with a slight northwest orientation, and Room 3 in the southwest corner. These rooms are accessed via a central landing next to a shared bathroom and stairwell. Room 1, on the eastern side, receives direct morning sunlight. It features double-glazed timber-framed casement windows with fabric curtains for manual daylight and thermal control. A radiator is positioned directly below the window, a standard but thermally inefficient setup due to convective heat loss through the glazing. Room 2 and Room 3 were not included in the current case study. The roof features an accessible loft, with sloped ceilings that contributes to variable ceiling heights and potential thermal stratification. The use of stone-masonry, double glazing, and conventional radiator systems forms the core of the building's thermal envelope. These architectural features are central to understanding microclimatic differences between rooms and validating sensor-based thermal comfort assessments. Focusing on this building, the study analyzes four key areas: Room 1 Window, Room 1, the Landing, and the Bathroom (see Fig. 4) to support building analysts in optimizing IECs.

These building areas were carefully selected to capture a range of temperature and humidity levels, as they significantly influence thermal comfort. Room 1 Window and Room 1 are crucial for assessing the differences between external environmental conditions and the indoor climate. The sensor at the window helps evaluate the effect of heat exchange with the outdoors, providing insights into how temperature fluctuations influence comfort throughout the day. Room 1, as the primary living area, represents the core of occupant experience and serves as a critical point for measuring indoor comfort. The sensor placed on the Landing monitors temperature and airflow in this shared, central part of the building, which can affect the comfort levels in adjacent rooms. As a transitional space, the Landing contributes to the building's thermal balance. Finally, the sensor in the Bathroom captures data in a space that often experiences unique thermal conditions due to its

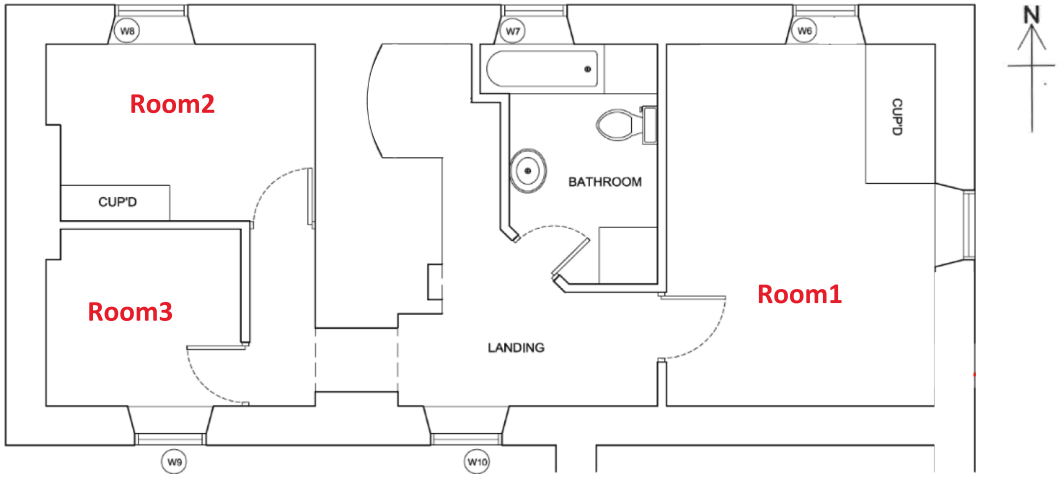


Fig. 4. Floor plan of the building for case study analysis.

frequent use, and the potential increased presence of moisture. This information is significant for understanding how temperature and humidity variations in this room affect overall comfort. By studying these specific locations, the objective is to identify any discrepancies in comfort across the building and develop data-driven solutions to ensure a consistently comfortable environment for the occupants.

4.2. Data acquisition and sensor placement strategy

An environmental monitoring system based on Zigbee technology is used to capture key environmental parameters, such as temperature and humidity. These wireless sensors communicate with a central monitoring hub, which stores the collected data locally on an internal Secure Digital (SD) card. The system can be accessed either locally through a built-in Wi-Fi connection or remotely via a secure Ethernet or Internet connection. Additionally, sensor data is automatically uploaded to an online server and visualized, enabling real-time monitoring and system configurations. To ensure data relevance and reliability, sensor placement within the indoor environment is strategically guided by an occupant-centric sensor positioning strategy [68]. This approach prioritizes placing sensors in zones where occupants spend most of their time, such as, living areas. Using this strategy, time-series temperature and humidity data were collected throughout the building over five-months (May to September). The acquired sensor dataset serves the basis for the case study presented in the following section.

4.3. Implementation of ThermalComfortBot in the case study.

In this case study, *ThermalComfortBot* serves as a comprehensive IS for integrating multiple data sources relevant to building TCM. It semantically combines BIM data (i.e., IFC file) with temperature and humidity sensor readings. This integration provides a holistic view of both the physical structure and the environmental conditions within the building. Additionally, *ThermalComfortBot* incorporates external weather data and HSE guidelines to ensure compliance with safety and comfort standards. Fig. 5 presents a systematic approach to analyzing temperature and humidity data for TCM using *ThermalComfortBot*. The analysis focuses on five key objectives from existing literature [7,12–19], aimed at optimizing occupant comfort by addressing the building's environmental conditions. The 1st objective compares temperature and humidity across locations to identify spatial variations and responses to external conditions. The 2nd detects anomalies in data,



Fig. 5. Systematic approach to TCM using *ThermalComfortBot*.

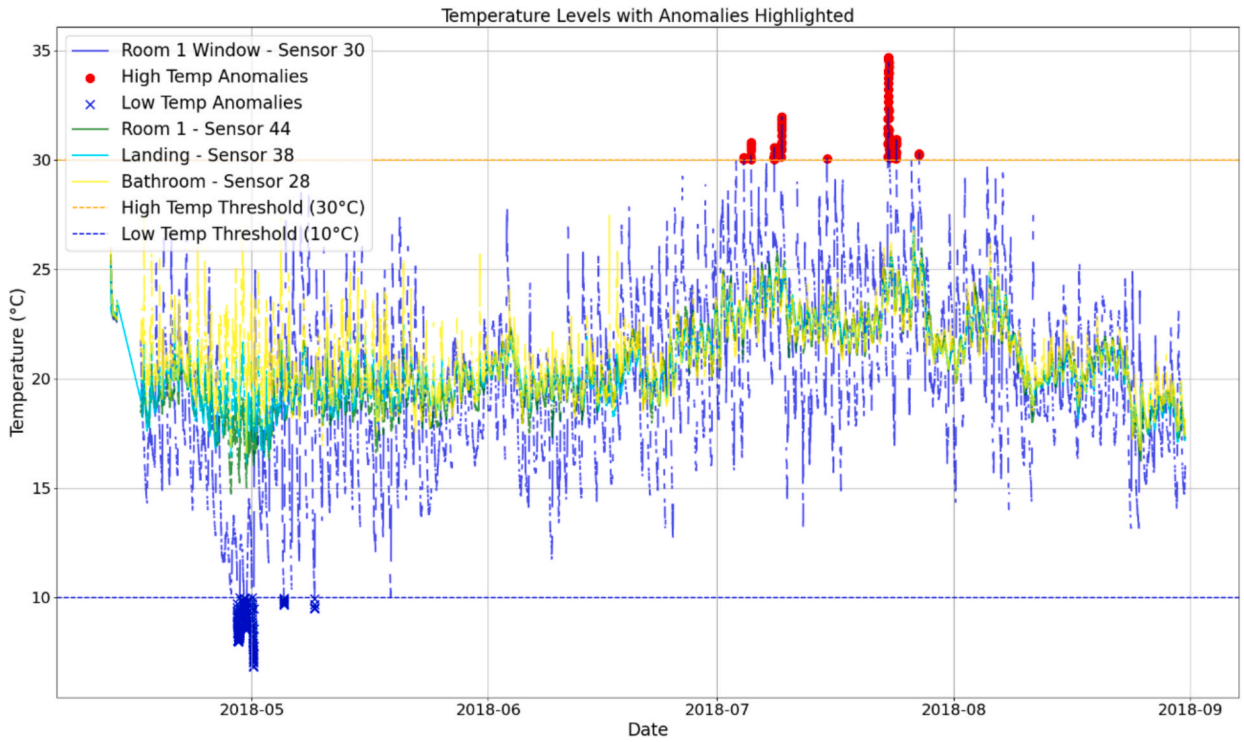


Fig. 6. Temperature anomalies in Room 1 Window, Room 1, Landing, and Bathroom areas within the building.

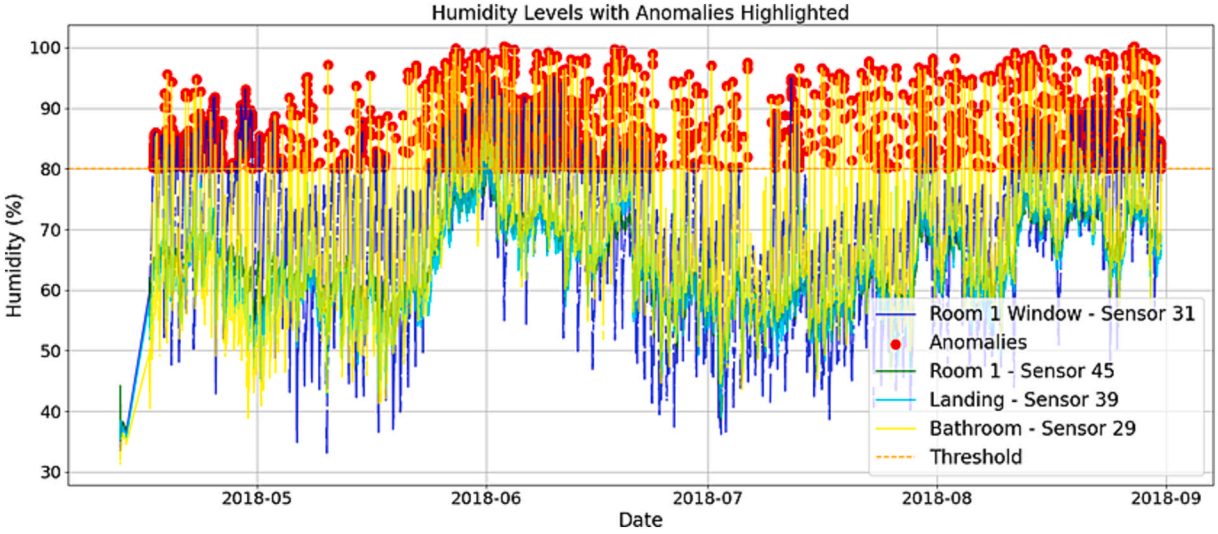


Fig. 7. Humidity anomalies in Room 1 Window, Room 1, Landing, and Bathroom areas within the building.

signalling potential Heating, Ventilation, and Air Conditioning (HVAC) issues. The 3rd objective investigates correlations between internal building environmental conditions and external climate conditions, which are crucial for improving energy efficiency and maintaining consistent comfort. The 4th objective monitors short and long-term variations in temperature and humidity by tracking fluctuations on a weekly, monthly, and seasonal basis, which is crucial for identifying periodic seasonal changes. Finally, the 5th objective focuses on understanding seasonal variations, mainly during the summer months when temperature and humidity levels naturally rise. Analyzing these sensor data fluctuations helps assess the capability of the building's cooling systems and determine whether further enhancements are necessary to maintain thermal comfort during periods of hot temperatures. Questions were formulated for each objective category based on these defined goals and input into *ThermalComfortBot* to assess its responses.

Table 1 presents a selection of user questions alongside the corresponding answers generated by *ThermalComfortBot*. Although the system is designed to provide both textual and graphical outputs, as outlined in Fig. 1, only a subset of questions from Fig. 5 are included for each objective. To maintain conciseness, graphical responses are not provided for every question.

4.4. Evaluation of the ThermalComfortBot

To evaluate the quality of the Chatbot's responses, focusing on their relevance, completeness, and the system's robustness, 50 queries were constructed based on the available datasets integrated into *ThermalComfortBot*. A representative subset of these queries is presented in

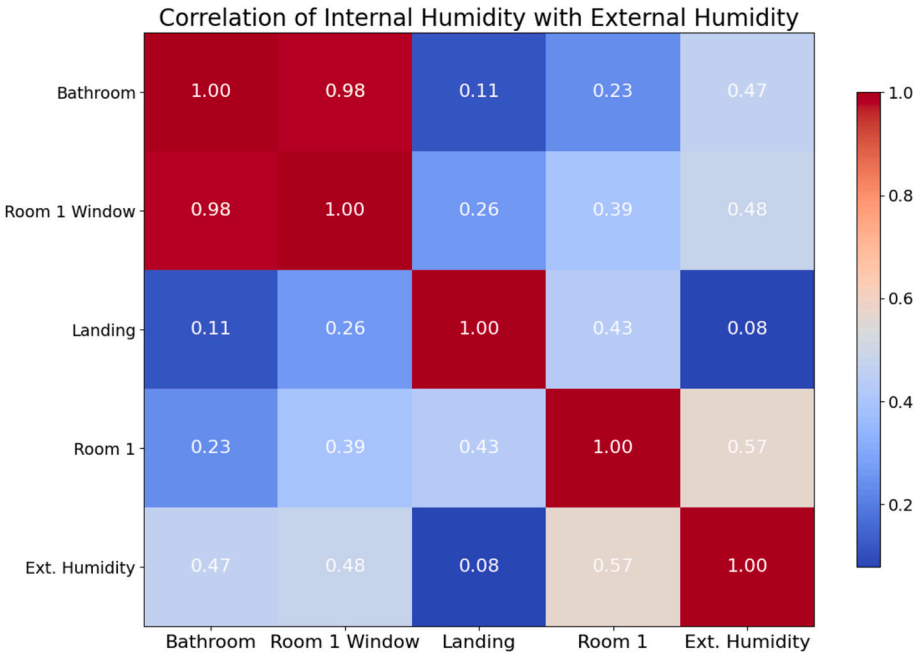


Fig. 8. Correlation between internal Humidity of Room 1 Window, Room 1, Landing, and Bathroom with External Humidity.

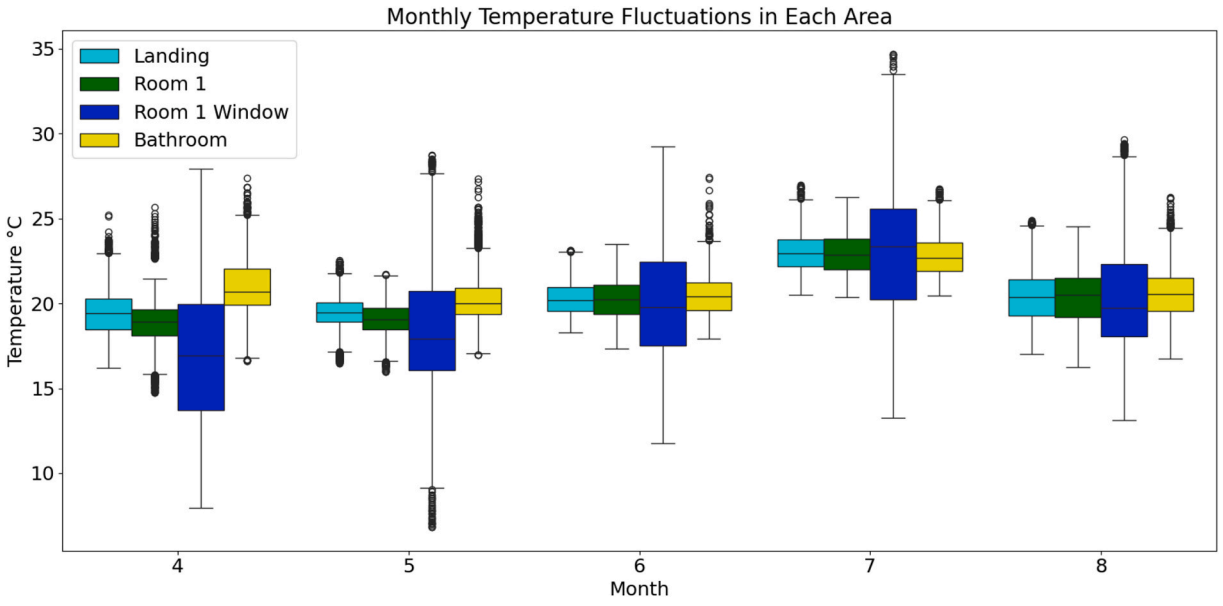


Fig. 9. Monthly Temperature fluctuations in Room 1 Window, Room 1, Landing, and Bathroom.

Table 2. These queries span various timeframes, locations, and types of data analysis to assess the system's capabilities thoroughly. The types of queries considered include:

1) *Time-Based Queries*: Requests for specific metrics (e.g., average or median temperature) over a defined period, such as "*What was the average temperature recorded in Room 1 Window during July?*" or "*How did the humidity levels in the Landing vary daily in May and June?*"
2) *Comparative Queries*: Comparing IECs across different spaces or timeframes, such as "*How do the average humidity levels in the Landing compare to those in the Bathroom in June?*" or "*What is the correlation between temperature data in the Landing and Room 1 during the summer?*"
3) *Extremes and Anomalies*: Identifying the highest, lowest, or outlier values in the data, such as "*Which sensor recorded the highest humidity levels in Room 1 during August?*" or "*What are the maximum and minimum temperatures recorded in Room 1 Window from May to September?*"
4) *Trend Analysis*: Analyzing patterns or trends over time, such as "*What were the weekly temperature trends in Room 1 Window during May?*" or "*What percentage of the days in August had humidity levels above 70 % in the Bathroom?*"
5) *Sensor-Based Queries*: Requests to compare data from different sensors or evaluate sensor performance, like "*How does the temperature recorded by a sensor in Room 1 differ from that in the Landing during July?*"

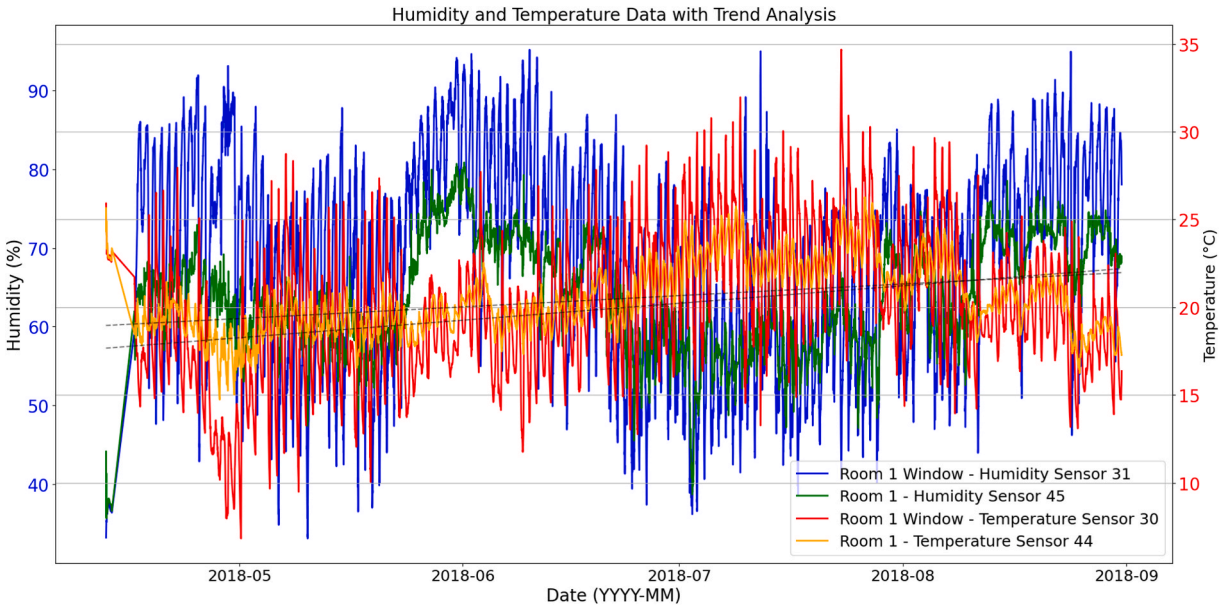


Fig. 10. Humidity and Temperature data from Room 1 Window and Room 1 for trend analysis.

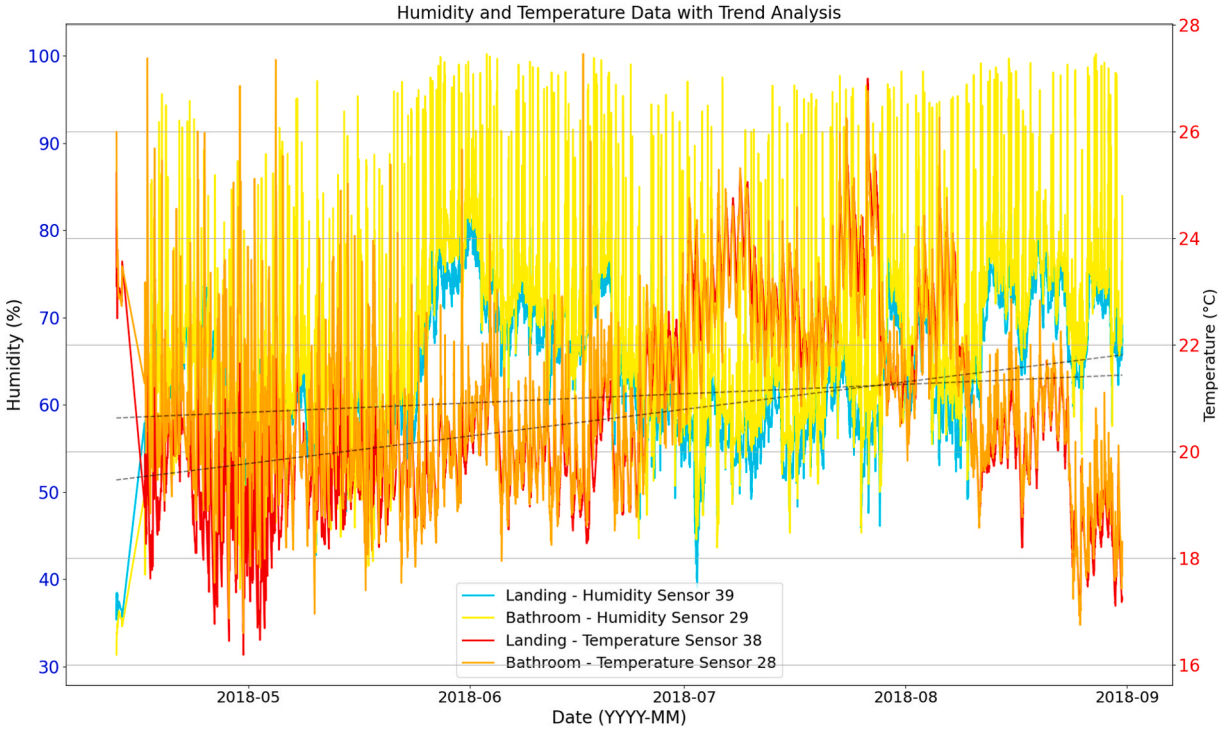


Fig. 11. Humidity and Temperature data from Landing and Bathroom for trend analysis.

These queries (see Fig. 5 and Table 2) evaluate the system's ability to handle complex requests, process diverse datasets (such as BIM and sensor data), and generate insights along with contextually relevant visualizations. To assess the responses generated by *ThermalComfortBot*, state-of-the-art evaluation metrics were used, including Accuracy, Precision, and Recall, which are widely employed in evaluating RAG-LLM-based systems [69].

These metrics are ideal for evaluating data-driven queries involving sensor readings, temperature, and humidity data, as they directly measure the system's ability to retrieve and present correct, relevant, and complete information. In contrast to evaluation metrics like BLEU or ROUGE [69,70], which focus on the fluency and grammatical quality of generated text, Accuracy, Precision, and Recall offer a more targeted assessment of the system's ability to deliver precise and contextually accurate responses. The primary goal was to evaluate the quality of the Chatbot's responses, emphasizing their relevance and completeness. To ensure a systematic evaluation, Information Components (ICs) [54,70,71] were defined, including sensor data values, building location names (e.g., Room 1 Window, Room 1, Landing, Bathroom), timestamps, sensor IDs, measurement units (e.g., Degrees Celsius for temperature, Percentage for humidity), and statistical terms (e.g., Average, Median). These ICs form the basis for evaluating *ThermalComfortBot's* capacity to provide accurate, relevant, and comprehensive information in response to user queries.

Table 1
Key questions on building thermal comfort and *ThermalComfortBot's* automated responses.

Objective	No.	Question	Answer
1) To compare temperature and humidity conditions across various monitored locations.	1	How does the temperature data from Room 1 Window compare to that of Room 1?	Room 1 Window (Sensor 30) experiences wider temperature fluctuations, with peaks reaching around 35 °C and lows dropping to approximately 10 °C, indicating higher variability. In contrast, Room 1 (Sensor 44) has more stable temperatures, ranging between 15 °C and 25 °C, with fewer extreme changes. While there are notable differences in variability, both sensors exhibit similar seasonal trends overall, though the Room 1 Window sensor appears more responsive to immediate environmental changes.
	2	How does the humidity data from the Landing compare to that of the Bathroom?	The bathroom shows significantly higher and more frequent humidity peaks, while the landing maintains a more moderate humidity range with fewer fluctuations.
2) To identify anomalies in temperature and humidity data.	3	What anomalies can be identified from the temperature data comparison of all areas?	**High-Temperature Anomalies:** The temperature readings exceed 30 °C, primarily occurring in July, indicating potential overheating. **General Observations:** **Key Anomalies:** High-temperature spikes in July 2018. Low-temperature drops in early May. See Fig. 6. **Low-Temperature Anomalies:** The temperature readings fall below 10 °C, mainly in early May, suggesting possible sensor malfunctions or actual cold conditions. • Significant fluctuations were noted across all sensors, with Room 1 Window (Sensor 30) showing the most pronounced high-temperature anomalies. • Sensors 28 (Bathroom) and 38 (Landing) generally remained within the expected range but occasionally reached higher thresholds.
	4	What anomalies can be identified from the humidity data comparison of all areas?	**High Humidity Anomalies:** Occur when humidity exceeds 80 %, mostly between mid-May and late August. **Early Anomalies**: Sparse from April to early May. **Peak Period**: Most anomalies from mid-May to late August, suggesting a seasonal influence. See Fig. 7. **Sensor-Specific Anomalies:** • **Room 1 Window (Sensor 31)**: Frequent anomalies consistently breaching the threshold. • **Room 1 (Sensor 45)**: Also shows many high-humidity anomalies. • **Landing (Sensor 39)**: Fewer anomalies but still exceeds the threshold occasionally. • **Bathroom (Sensor 29)**: Exhibits few anomalies, indicating more stable humidity. **Temporal Distribution:**
3) To assess how conditions in different areas are related, helping to understand spatial dependencies.	5	How do temperature levels correlate between different areas?	The Landing and Bathroom areas show the strongest correlation in temperature, with a coefficient of 0.97, indicating nearly identical temperature changes. Room 1 Window also has high correlations with both Landing (0.86) and Bathroom (0.81), though it correlates less with Room 1, at 0.43. Room 1 itself demonstrates only moderate correlations with other areas, suggesting more distinct temperature patterns that are less influenced by changes in the Landing or Bathroom.
	6	How does internal humidity correlate with external humidity?	Room 1 and Room 1 Window are most influenced by external humidity, followed by the Bathroom, while the Landing is the least affected. Overall, internal humidity in some rooms is moderately impacted by external humidity, while others remain relatively isolated. See Fig. 8.
4) To monitor short and long-term variations in environmental conditions for trend analysis.	7	What are the weekly humidity fluctuations in each room?	**Landing:** Humidity varies between 50 % and 70 %, with increased variability in weeks like 21 and 29. Drops to around 40 %

(*continued on next page*)

Table 1 (*continued*)

Objective	No.	Question	Answer
			occur in weeks 19 and 27, with more frequent outliers. In short, Room 1 Window and the Bathroom have the highest humidity fluctuations, often nearing 90 %, while Room 1 is the most stable and the Landing shows moderate levels with occasional drops. **Room 1:** Typically stable at 60 % to 80 %, with slight dips in weeks 20 and 25. Shows tighter variability, indicating a more controlled environment. **Bathroom:** Experiences higher fluctuations, ranging from 60 % to 90 %. Notable spreads in weeks 17, 19, and 22, with drops below 60 % and peaks around 80 % to 90 %. More outliers are seen in weeks 23, 27, and 32. **Room 1 Window:** The most variable, with humidity levels spanning from 50 % to 90 %. Frequently reaches high peaks (80 %-90 %) in weeks 16, 20, and 24, indicating greater exposure to external conditions.
	8	What are the monthly temperature fluctuations in each room?	• **Room 1** maintains a temperature range of 17–23 °C with a median around 20 °C, showing slight fluctuations throughout the months. In short, **Room 1 Window** is the coolest and most variable, while the **Bathroom** is the warmest and most stable. **Room 1** and **Landing** show moderate temperatures, generally warmer than the window area but cooler than the bathroom. See Fig. 9. • **Room 1 Window** starts cooler, with temperatures ranging from 13-22 °C in April and a median of 16 °C. It fluctuates more significantly, reaching a range of 14–23 °C in July, before stabilizing at 13–22 °C in August. • **Landing** has temperatures ranging from 15-23 °C in April, with a median of 19 °C, and remains consistent throughout the months, usually between 17–22 °C. • **Bathroom** maintains the warmest temperatures, ranging from 19-24 °C in April with a median of 21 °C, remaining stable at around 21 °C through August.
5) To understand the effects of seasonal changes on temperature and humidity in various areas.	9	How do humidity and temperature levels in Room 1 Window and Room 1 areas vary throughout the summer?	**Humidity Trends:** **Temperature Trends:** In short, both Room 1 and the window show significant humidity fluctuations, with the window experiencing more extreme variations. Temperature peaks occur in July and August, with the window recording higher temperatures. Both areas display a cooling trend approaching September. See Fig. 10. • **Room 1 Window (Sensor 31)**: Humidity fluctuates significantly, with frequent spikes above 80 % during summer. While it dips to 50–60 % at times, the overall pattern is erratic, showing a slight decreasing trend toward August-September. • **Room 1 (Sensor 45)**: Indoor humidity fluctuates less dramatically, mainly staying between 50–70 %. A slight decreasing trend suggests the room may be getting drier as summer ends. • **Room 1 Window (Sensor 30)**: Temperatures range from 15 °C to 35 °C, with peaks of 30–35 °C in July and August after lower readings in May and June. A slight cooling trend is noted toward September. • **Room 1 (Sensor 44)**: Indoor temperatures are more stable, ranging from 15 °C to 25 °C, with peaks around 25 °C in July and August. A slight cooling trend is also evident as summer concludes.

(*continued on next page*)

Table 1 (*continued*)

Objective	No.	Question	Answer
	10	How do humidity and temperature levels in the landing and bathroom areas vary throughout the summer?	**Humidity Trends: Temperature Trends:** In short, the bathroom has higher humidity, more fluctuating humidity levels than the landing, with both areas trending upward throughout summer. Both areas warm through the summer, with bathroom temperatures slightly higher at peaks, ranging from 16 °C to 26 °C. See Fig. 11. • **Landing (Sensor 39)**: Humidity fluctuates between 40 % and 80 %, with sharp rises and drops in summer. From mid-July to August, it stabilizes above 60 %, showing a slight increasing trend overall. • **Bathroom (Sensor 29)**: Bathroom humidity is highly variable, often exceeding 90 %, indicating constant moisture. Levels range from 50 % to 100 %, with a slight increasing trend by summer's end. • **Landing (Sensor 38)**: Temperature ranges from 18 °C to 24 °C, remaining steady with peaks in July and early August. There's a slight increase over the summer before cooling in September. • **Bathroom (Sensor 28)**: Bathroom temperatures vary from 16 °C to 26 °C, peaking in July and August. The bathroom shows slightly more variation but follows a similar upward trend as the landing.

Information Accuracy [70] refers to how closely the Chatbot's responses align with the correct answers across all questions, considering all ICs. It is the ratio of the number of correctly provided answers (having correct data values, units, and statistics) to the total number of responses generated. Precision measures the correctness of the Chatbot's responses, specifically focusing on how relevant and accurate the provided information is. In this context, Precision [70] would be evaluated by checking if the Chatbot correctly matches the ICs. It is calculated as the ratio of correct information to the total information provided in the Chatbot's response. Recall evaluates the completeness of the Chatbot's responses. In terms of ICs, Recall measures whether the Chatbot has included all the necessary ICs to answer the user's question completely.

Table 2
Test Questions for Evaluating *Thermalcomfortbot* Performance.

No.	Test Questions
1	What was the average temperature recorded in Room 1 Window during July?
2	How did the humidity levels in the Landing vary daily in May and June?
3	What was the median temperature in the Bathroom in September?
4	Which sensor recorded the highest humidity levels in Room 1 during August?
5	What are the maximum and minimum temperatures recorded in Room 1 Window from May to September?
6	How do the average humidity levels in the Landing compare to those in the Bathroom in June?
7	What were the weekly temperature trends in Room 1 Window during May?
8	How does the temperature recorded by a sensor in Room 1 differ from that in the Landing during July?
9	What percentage of the days in August had humidity levels above 70 % in the Bathroom?
10	What is the correlation between temperature data in the Landing and Room 1 during the summer?

For instance, if the question asks about "*temperature variations over a specific period*", Recall will ensure the Chatbot provides not only the correct temperature values but also the associated timestamp, sensor ID, and statistical summary (e.g., Average or Median). Recall [70] is calculated by dividing the number of correct, complete responses by the total number of correct, required ICs that should have been provided.

Table 3 highlights the superior performance of *ThermalComfortBot* when integrated with the LLaMA 3.1 model and Agentic RAG. The system achieves 94 % Accuracy, indicating that most generated responses align closely with ground truth answers. Moreover, the system gives a Precision of 92 %, demonstrating its ability to produce responses that are not only correct but also highly relevant to user queries. The Recall score of 89 % further highlights the system's effectiveness in retrieving most of the necessary information across diverse data sources. To evaluate the added value of the Agentic RAG mechanism, its performance is directly compared with a traditional RAG-LLM setup using the same LLaMA 3.1 model. The results show a significant performance gap. The standard RAG setup lags in Accuracy (82 %), Precision (85 %), and Recall (81 %). This comparison clearly illustrates that the Agentic RAG framework enhances IR and reasoning by enabling autonomous agents to plan, fetch, and integrate knowledge from heterogeneous datasets. As a result, the Agentic RAG-driven system provides more contextually accurate and complete responses, which are critical in the domain of TCM.

While *ThermalComfortBot* exhibits strong overall performance, a closer analysis of complex queries (as shown in Table 2) highlights key limitations of traditional RAG-LLM systems, particularly in scenarios that demand multistep reasoning and integration across diverse data sources. Queries involving comparative analysis, temporal alignment, or spatial mapping, such as "*How do the average humidity levels in the Landing compare to those in the Bathroom in June?*" or "*What is the correlation between temperature data in the Landing and Room 1 during the summer?*", were not accurately answered by the standard RAG-LLM model. These types of queries require more than basic IR. These queries involve structured reasoning across various heterogeneous datasets (e.g., time series sensor logs, BIM models), a capability that conventional RAG architectures are not fundamentally built to support. Performance failures in these cases are often caused by misaligned IR,

Table 3
Comparative Performance of *Thermalcomfortbot* Using LLAMA 3.1 with Agentic VS. Standard RAG Integration.

Model Configuration	Accuracy (%)	Precision (%)	Recall (%)
ThermalComfortBot (Llama 3.1 & Agentic RAG)	94	92	89
ThermalComfortBot (Llama 3.1 & Standard RAG)	82	85	81

ambiguous sensor references, or the LLM's limited ability to execute logical inference. For example, if humidity data retrieved from two rooms is not temporally synchronized, the answer lacks contextual validity, leading to lower Precision and Recall. Similarly, if sensor IDs are not clearly mapped to building locations, the response may become misleading. This is where Agentic RAG architecture shows a significant advantage. By introducing autonomous planning and tool orchestration, Agentic RAG enables the system to decompose complex queries into subtasks, utilize domain-specific retrievers (e.g., for time series sensor data analysis and BIM interpretation), and synthesize the outputs before passing them to the LLM [55,56]. This goal-directed and modular behavior significantly enhances both IR reliability and the consistency of the final LLM's response, specifically in multi-hop reasoning tasks. These findings strengthen that Agentic RAG is not only suited for performance optimization, but necessary architectural advancement for domains like facilities management, where queries often span structured and semi-structured data and require context-aware synthesis rather than isolated fact-based IR.

5. Discussion

This study presents *ThermalComfortBot*, an IS designed to improve building TCM by integrating various datasets, including building and sensor data. Tested in a real-world case study, the system's accuracy and effectiveness were evaluated against five key objectives (see Fig. 5), which address critical aspects of IEC monitoring. These objectives help assess thermal comfort through spatial, temporal, and analytical insights, enabling more precise control of HVAC systems, anomaly detection, correlation analysis, and trend analysis. Room-to-room comparisons, for example, highlight localized temperature and humidity variations, supporting better thermal design decisions and energy-efficient planning. The system's ability to automate these analyses provides a comprehensive understanding of thermal dynamics, driving optimized building management [71]. Table 4 compares the benefits of the proposed system with existing thermal monitoring systems [28–34] across several key aspects: *Data Handling*, *User Interaction*, *Query Response Time*, *Objective Coverage*, *Visualization Support*, *Accuracy*, *Scalability & Sustainability*, *Insights and Reasoning*, *Fault Detection, and Accessibility*. These factors highlight how *ThermalComfortBot* improves efficiency, accuracy, and decision-making capabilities in building TCM.

The outcomes generated by the *ThermalComfortBot* can play a vital role in preventing health hazards in buildings by ensuring optimal thermal comfort. By continuously monitoring and analyzing temperature and humidity across different building zones, the system identifies potential risks associated with extreme environmental conditions, including heat stress, inadequate ventilation, or excessive moisture. For example, the system can prompt immediate corrective actions, such as adjusting HVAC settings or improving airflow, in areas where temperature or humidity exceeds safe thresholds. Additionally, by providing real-time data and trend analysis, the Chatbot helps detect early signs of building-related health issues, such as respiratory problems, mold growth, or heat-related illnesses. These proactive insights enable building managers to make informed decisions that not only enhance occupant comfort but also prevent exposure to environmental factors that could lead to long-term health problems. Despite the many benefits of the *ThermalComfortBot*, several limitations remain that should be addressed in future work.

1. *System's Fault Detection Mechanisms*: This study does not address the strategies for handling anomalies or missing values in sensor data. Although the system can detect and highlight such issues using LLMs, it is not currently equipped to correct them. For example, although missing or inconsistent data points can be identified, the system does not apply corrective techniques, such as interpolating values between preceding and succeeding sensor readings. Integrating such capabilities would enhance the system's overall reliability and effectiveness in real-world building management scenarios.
2. *Lack of Contextual Details*: Although the study employed an office building case study, specific details, such as building type, occupancy numbers, and other relevant contextual factors, were omitted due to concerns about data confidentiality. However, the data integration approach remains applicable across different building types and occupancy scenarios, making the methodology versatile.
3. *Limited Case Study Scope:* The current case study was intentionally designed to cover a restricted area of the building, primarily serving as a proof-of-concept to validate the feasibility and performance of the proposed *ThermalComfortBot* system. Although this approach effectively demonstrates the system's core functionality, it does not fully capture the variability and complexity of thermal comfort dynamics across diverse building zones or different building types. A broader implementation, spanning multiple rooms, floors, or even other buildings, would provide a more comprehensive understanding of how thermal comfort is affected by various factors such as occupancy patterns, HVAC configurations, material properties, and

Table 4
Comparison of State-of-the-Art TCM Systems vs. *ThermalComfortBot*.

No.	Aspect	State-of-the-Art Systems	*ThermalComfortBot* (Proposed System)
1	Data Handling	Manual integration of BIM, sensor, and textual data.	Automated multi-source data fusion via Agentic RAG framework.
2	User Interaction	Technical expertise required. Use of scripts, spreadsheets, or dashboards.	NL interface via Chatbot. No coding or manual queries needed.
3	Query Response Time	Data collection and processing make time-consuming processes.	Instantaneous responses with automated IR and reasoning.
4	Objective Coverage	Requires manual setup for each analytical goal.	Predefined analytical objectives (e.g., trends, anomalies, correlations) are built-in.
5	Visualization Support	Requires manual plotting and dashboard configuration.	Auto-generated visual outputs (e.g., time-series plots, comparisons, etc.).
6	Accuracy	Varies with analyst expertise and prone to human error.	Consistently high accuracy (94 %), precision (92 %), and recall (89 %) from evaluation.
7	Scalability & Sustainability	Limited scalability to new datasets and buildings. Frequent manual updates and a higher carbon footprint due to inefficiencies.	Highly scalable across buildings with diverse structures and configurations. The system accommodates evolving datasets without requiring reengineering. Using the RAG framework, only updated data files are needed, allowing quick replacements without technical expertise. It also promotes sustainability through energy-efficient monitoring and low technical overhead, making it ideal for dynamic and adaptive building environments.
8	Insights and Reasoning	Domain-specific systems require human interpretation.	Deep reasoning through LLM and RAG supports statistical and semantic insights.
9	Fault Detection	Relies on manual observation or predefined thresholds.	Automatic anomaly detection through learned patterns and contextual understanding.
10	Accessibility	Accessibility is limited to experts in data analysis.	Usable by facility managers, building analysts, or non-technical staff.

environmental zones. Such an expansion would not only enhance the generalizability of the findings but also help in refining the system's adaptability to real-world conditions. However, scaling up the system by incorporating more buildings and a larger volume of sensor data may significantly affect data processing and retrieval performance. In particular, the query response times and the efficiency of the RAG mechanism may be affected by the increased data complexity and volume. This trade-off between system scalability and responsiveness must be carefully evaluated in future work.

4. *Sensor Calibration*: Regular calibration of environmental sensors is essential to ensure the long-term accuracy and reliability of the collected data. Although the study acknowledges the importance of sensor calibration, it does not provide detailed strategies for optimal calibration in the context of varying building environments. Occupancy density, usage patterns, and seasonal changes can significantly influence sensor performance and contribute to data drift. A more comprehensive approach that addresses calibration frequency, environmental influences, and real-time validation methods would enhance the system's robustness and improve the reliability of its thermal comfort assessments.
5. *Limited Use of BIM Data*: BIM data was primarily used to provide spatial context for thermal analysis, helping *ThermalComfortBot* to understand the building's layout. While simpler Computer-Aided Design (CAD) files could have sufficed for this purpose, a deeper integration of enriched BIM data could significantly enhance the decision-making capabilities of the system. The system could enhance thermal comfort assessments by incorporating factors such as interactions between environmental conditions, occupancy, HVAC systems, and external weather. Additionally, BIM could play a key role in optimizing HVAC systems and predicting heat transfer, leading to more efficient thermal and energy management.
6. *Single LLM Implementation*: The system was implemented with a single LLM from the Llama family, chosen for its open-source availability. Although this LLM proved effective for initial testing, future research could explore other LLMs, such as those from the GPT family, to compare performance and enhance the system's capabilities.
7. *Lack of Real-World Feedback*: The system was evaluated based on Precision, Recall, and Accuracy from user queries, but feedback from building analysts and facility managers was not incorporated. Real-world feedback is essential for assessing the system's practical usability and incorporating this can provide valuable insights into the system's operational success in actual building management scenarios.
8. *Bias and Accuracy in Multi-Source Data Integration:* A limitation of the proposed system is the potential for bias or inaccuracy in data retrieval and synthesis, particularly when dealing with multi-dimensional queries involving real-time sensor inputs (e.g., temperature, humidity) and static data (e.g., BIM metadata). In RAG-based frameworks, retrieved content may not always align perfectly with the specific query's contextual needs, which can affect the relevance and accuracy of the LLM-generated outputs. To alleviate this, domain-specific filtering techniques were applied during IR and the query pipeline was designed to align more closely with the semantic structure of the source data. While these steps help reduce inconsistencies, biases may still arise due to limitations in the underlying datasets, retrieval errors, or inference behavior of the language model. Future work will focus on incorporating bias detection mechanisms, provenance tracking, and confidence scoring to improve the robustness and trustworthiness of generated insights in dynamic building environments.

These limitations highlight areas for future research to enhance the system's effectiveness and suitability in real-world settings. Despite these limitations, *ThermalComfortBot* offers a practical approach for organizations seeking to enhance indoor environmental monitoring with minimal R&D investment. Its modular design and open architecture make it highly adaptable, allowing organizations to implement and customize the system according to their specific operational requirements and infrastructure. This work lays a solid foundation for future research, particularly in the integration of emerging technologies such as Augmented Reality (AR) and advanced voice-to-text services [72]. Adding voice-to-text functionality would significantly enhance the accessibility and usability of the system, enabling building analysts, facility managers, or maintenance staff to interact with the system through natural spoken language rather than relying solely on typed input. This hands-free interaction is particularly beneficial in dynamic environments or during on-site inspections. Moreover, AR integration could further enhance the user experience by enabling real-time visualization of environmental data through dynamic heatmaps, alerts, and contextual overlays. These visual enhancements can be projected onto physical spaces using AR headsets or mobile devices, offering intuitive and spatially relevant feedback. Such advancements would streamline data interpretation and facilitate faster and more informed decision-making in facility management, energy optimization, and occupant comfort analysis.

6. Conclusion

ThermalComfortBot is a purpose-built IS for facility management. It is designed to enhance thermal comfort, providing direct benefits for occupant health, operational efficiency, and energy management. As the first Agentic RAG-LLM-based solution to integrate structured, semi-structured, and unstructured data for TCM, it lacks direct benchmarks in the existing literature. However, real-world validation showed strong performance, achieving 94 % Accuracy, 92 % Precision, and 89 % Recall. *ThermalComfortBot* offers several advantages over existing thermal monitoring systems. It provides an NL interface that allows users to interact with the system without requiring any technical expertise in data analysis. Unlike existing RAG-LLM systems, it also offers graphical outputs to support faster data interpretation. It delivers context-aware insights by integrating multi-format data. The open-source, easy-to-deploy, and scalable architecture of the proposed system further augments its practicality. Its cost-efficiency and alignment with sustainability goals position it as a forward-looking and innovative solution in building management. The system supports smarter energy use by enabling effective TCM, contributing directly to the development of energy-efficient and environmentally responsible buildings. *ThermalComfortBot* offers a practical foundation that organizations can adopt with minimal R&D effort, despite its limitations. Future work could explore the integration of AR and voice-to-text services to enhance usability and real-time decision-making by enabling voice-based interaction and dynamic visual overlays, such as heatmaps.

CRediT authorship contribution statement

Muhammad Arslan: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Saba Munawar:** Writing – review & editing, Software, Methodology. **Lamine Mahdjoubi:** Writing – review & editing, Data curation. **Patrick Manu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies

As this study incorporates the use of Large Language Models (LLMs), the answers presented in Table 1 were generated using the LLaMA 3.1 model. These examples demonstrate the model's ability to interpret and generate contextually relevant outputs based on a private dataset, highlighting its value within the study framework. Additionally, GPT-4-turbo model is used to facilitate proof-reading and clarity of the writing of this paper.

Data availability

Data will be made available on request.

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